



AHLs' life in plants: Especially their potential roles in responding to *Fusarium* wilt and repressing the seed oil accumulation

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ABSTRACT

Members of the AT-hook motif nuclear localized (AHL) family contain diverse but poorly understood biological functions. We identified 371 AHLs in 20 land plants, varying from the early diverging lycophyte *Selaginella moellendorfi* to a variety of higher plants. The AHLs were divided into two clades (Clade-A and Clade-B) with three different types (Type-I, Type-II, and Type-III AHLs). The divergence between Clade-A and Clade-B likely occurred before the separation of *S. moellendorfi* from the vascular plant lineages. Members of the AHLs family expanded with the specific whole-genome duplication (WGD)/segmental duplication in some genomes, such as *Hevea brasiliensis*. The ortholog (Vf00G1914/Amo018442) exhibited opposite expression patterns between two *Vernicia* species (*V. fordii* and *V. montana*), indicating that it was implicated in resistance to *Fusarium* wilt disease. The expression of Vf09G2138 exhibited a negative correlation with lipid biosynthesis in *V. fordii* seeds during different stages of development, suggesting that this gene might repress the seed oil accumulation. The core AT-hook motif and PPC domain were responsible for guiding the localization of AHL in the nucleus. This study helps us to understand the evolution of AHLs in multiple plants, further highlight their functions during *V. fordii* seed development and response to *Fusarium* wilt disease.

1. Introduction

The ancient AT-Hook Motif Containing Nuclear Localized (AHL), which serves as a conserved transcription factor, exists in all sequenced land plants, varying from the lycophyte *Selaginella moellendorfi* to higher plants, such as *Vernicia fordii*, *Oryza sativa*, *Vitis vinifera*, and *Arabidopsis thaliana*. Members of the AHL gene family contain two domains, the plants and prokaryotes conserved (PPC) domain and the AT-hook motif [1]. The PPC domain contains approximately 120 amino acids that might regulate the transcriptional activation of AHL proteins [2]. The AT-hook motif contains a conserved palindromic core sequence (i.e. Arg-Gly-Arg) binding to the AT-rich DNA region [1,3,4]. According to the number and type of AT-hook motif(s), the AHLs can be divided into

three types: Type-I, Type-II, and Type-III [3,5–7].

The AHL family has been identified in some plant species, such as *Zea mays*, *Populus trichocarpa*, *O. sativa*, and *A. thaliana* [1,5,6,8,9]. The systematic analysis of AHLs suggests that these genes are essential in regulating plant growth and development. In *A. thaliana*, several AHLs, such as AtAHL6, AtAHL15, AtAHL22, AtAHL27, and AtAHL29, play key roles in regulating the elongation of hypocotyl [1,4,10,11]. AHLs are associated with the reproductive development and floral transition of plants [4,12–15]. Overexpression of AtAHL20 in *A. thaliana* will result in a decrease in *FT* expression and finally delayed flowering [15]. Overexpression of AtAHL18 in *A. thaliana* can promote root growth, making it have higher lateral root density and longer primary roots [16]. AtAHL22 acts as a chromatin remodeling factor by recruiting histone deacetylases

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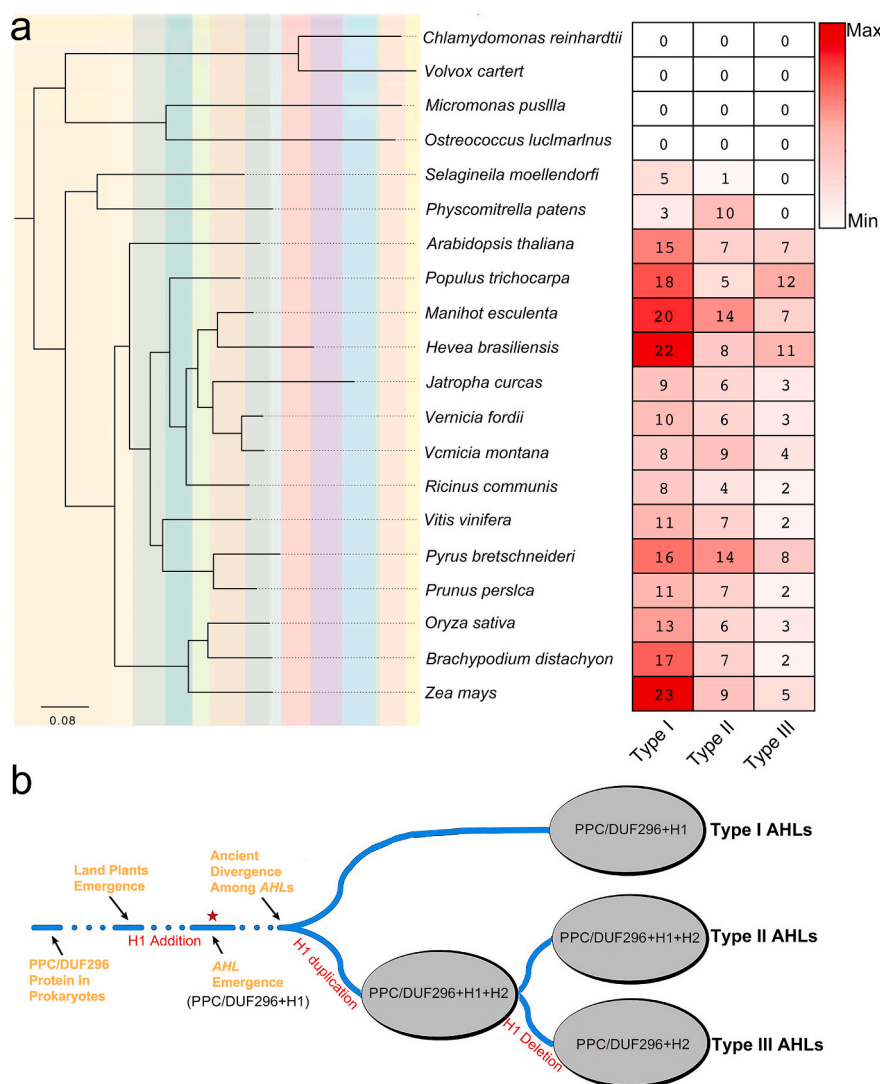


Fig. 1. Species tree of 20 plants and AHL numbers of each species. (a) The Orthofinder was used to detect one-to-one orthologs between genomes [51], and the IQ-tree was used to construct species tree (left) [25]. The numbers of AHL family detected in each species were listed accordingly (right). (b) A hypothetic model for AHLs evolutionary trajectory in land plants. H1 and H2 suggested AT-hook motif1 and AT-hook motif2, respectively.

(HDAC) and then delays flowering [12]. The *GIANT KILLER* (*GIK*) coding an AT-hook protein leads to down-regulation of genes related to the differentiation of reproductive floral organs, such as *KNUCKLES* (*KNU*), *JAGGED* (*JAG*), and *ETTIN* (*ETT*) genes [17]. In addition, members of the AHL family also play important roles in regulating defense response and maintaining hormones homeostasis [18,19]. In *P. trichocarpa*, *PtAHL8*, *PtAHL9*, *PtAHL16*, *PtAHL20*, *PtAHL22*, *PtAHL23*, *PtAHL27*, *PtAHL29*, *PtAHL34*, and *PtAHL36* are induced by drought stress in both leaves and roots [5]. Overexpression of *OsAHL1* can greatly improve the drought tolerance and drought avoidance in *O. sativa* [19].

In the present study, we carried out a genome-wide analysis of AHLs in 20 genomes using the latest version of the genome, which aims to identify AHLs and investigate family evolutionary patterns, gene duplication types, and expression patterns in the genes of different tissues and developing seeds and flower of *V. fordii*. Additionally, the *Vernicia* trees, *V. fordii* and *V. montana*, are two primary cultivated plants in China. Compared with *V. montana*, *V. fordii* exhibits superior oil characteristics and faster maturation [20]. However, *V. fordii* was found to be extremely susceptible to *Fusarium* wilt disease, while *V. montana* exhibited significant resistance to this disease [21]. Therefore, the transcriptional activity of orthologous AHLs was also compared after

Fusarium wilt infection between *V. montana* and *V. fordii*, to further understand the molecular mechanism of *V. montana* resistance to *Fusarium* wilt. Considering the high conservation of AHLs, the goal of our study is not only to provide a further understanding of this family, but also to survey the genome characteristics and the expansion of gene evolution in plants.

2. Materials and methods

2.1. Data retrieval

Firstly, the genomes of *P. trichocarpa*, *Brachypodium distachyon*, *V. vinifera*, *Prunus persica*, *Z. mays*, *Ricinus communis*, *Manihot esculenta*, *O. sativa*, *Physcomitrella patens*, *S. moellendorfi*, *Micromonas pusilla*, *Ostreococcus lucimarinus*, *Chlamydomonas reinhardtii*, and *Volvox carter* were obtained from Phytozome v13 database (<https://phytozome-next.jgi.doe.gov/>). Both *V. fordii* and *Hevea brasiliensis* genomes were downloaded from NCBI database (<https://www.ncbi.nlm.nih.gov/>). The genome of *A. thaliana* was obtained from TAIR database (<https://www.arabidopsis.org/>), and the genome of *Pyrus bretschneideri* was downloaded from GigaDB (<http://gigadb.org/>). Subsequently, the Hidden Markov Model (HMM) of PPC/DUF296 (PF03479) in the Pfam database

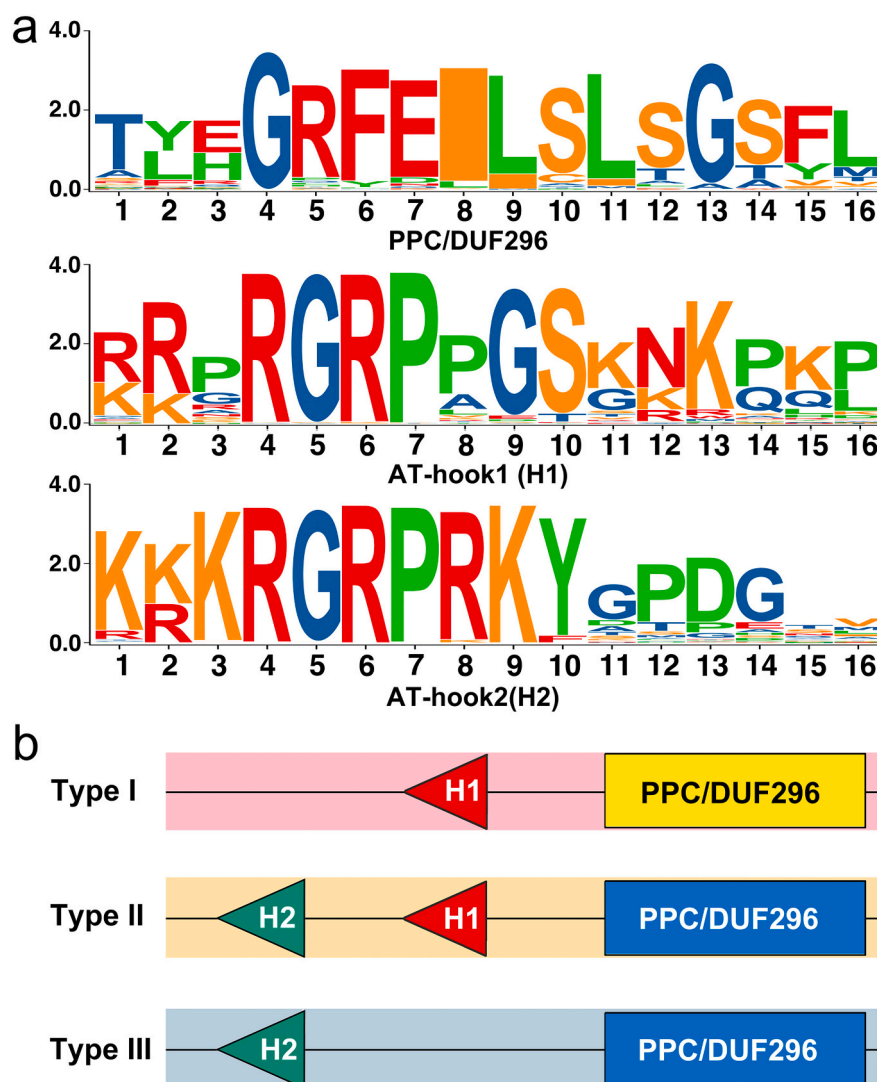


Fig. 2. Domain composition of AHLs in land plants. (a) The conserved motifs of H1, H2 and PPC which obtained from MEME [52]. The x-axis and y-axis suggested the conserved sequences of the domain and the conservation rate of each amino acid, respectively. (b) The topology of three AHLs types (Type-I AHLs, Type-II AHLs, and Type-III AHLs) determined in land plants according to the PPC domain and AT-hook motif(s).

was used for gene search among twenty genomes, as described by Cao et al. (2020) [22]. The amino acid sequences of *A. thaliana* AtAHLs were further used as queries to identify AHLs in the local genome database using BLASTp with an E-value of $1e-5$. The obtained AHLs were further confirmed by searching against the NCBI CDD database with an E-value of $1e-5$ [23]. Finally, the unique sequences only contained both PPC domain and AT-hook motif(s) were selected as putative AHLs.

2.2. Phylogenetic and duplication analysis

MAFFT software was used to align the amino acid sequences of the Type-I AHLs, Type-II AHLs and Type-III AHLs, respectively [24]. According to the best model as implemented in IQ-tree, the maximum likelihood (ML) trees were computed and built with a bootstrap test (5000 replicates) and an SH-aLRT test (1000 random addition replicates) [25]. Both Figtree and iTOL were used to visualize ML trees [26]. The MCScanX was used to carry out the collinearity analyses of all AHLs between and within species with an E-value of $1e-5$ [27]. Both TBtools and Circos were used to visualize the collinear relationships of AHLs and the distribution of these genes on the chromosomes [28,29].

2.3. Gene expression pattern

To gain the expression patterns of *VfAHLs*, we obtained and downloaded the RNA-seq data from SRA database with accession numbers: PRJNA318350, PRJNA483508, and PRJNA445068. Then we used the trimmomatic to delete and remove the low-quality reads and sequencing adapters. The clean reads were mapped into its corresponding reference genome by HISAT2 with default parameters [30]. The counts of reads were calculated by StringTie with default parameters, and FPKM was used to normalize these expression data, as described by Cao et al. (2019) [31].

2.4. Site-directed mutagenesis of *Vf09G2138* (AHLmut) by PCR and subcellular localization

Firstly, we designed a pair of reverse complement long primers containing mutation sites (RGRP to AAAA), *Vf09G2138*-mut F/R. Using the cDNA of *Vf09G2138* as a template, the partial sequence of the *Vf09G2138* was amplified by the primers of AHLaR/AHL-mutF and AHLaF/AHL-mutR, respectively. Then we used the mixture of the above products as a template to amplify the complete sequence using primers AHLaF/R. Finally, most of the amplified products were mutant

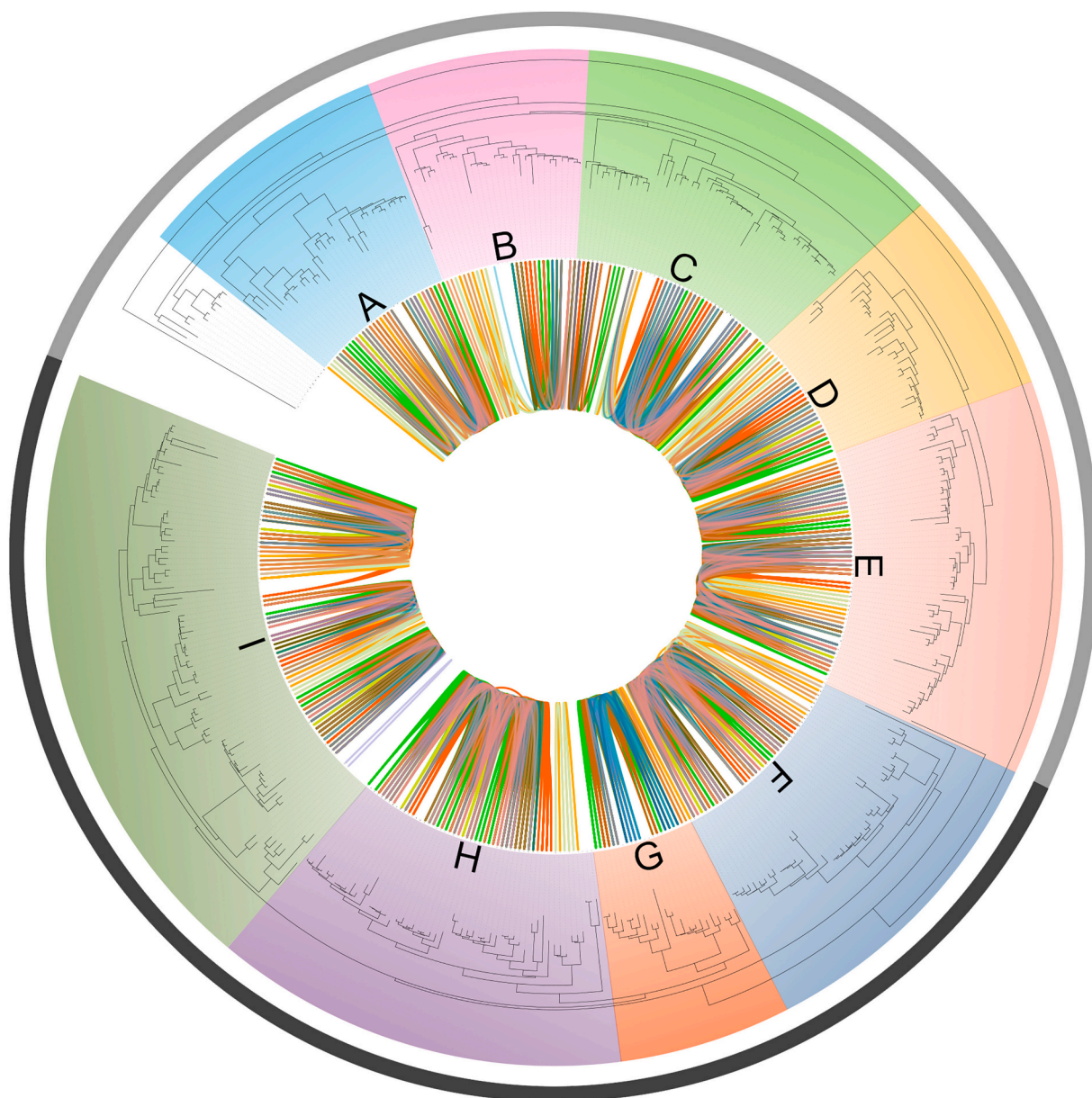


Fig. 3. Analysis of maximum likelihood (ML) tree and orthologous gene pairs among the land plants. Gray and black represented Clade-A and Clade-B, respectively. The AHLs of Clade-A could be divided into five subfamilies, including A, B, C, D and E. The AHLs of Clade-B are distinguished into four subfamilies, including F, G, H and I. The different color links suggested orthologous gene relationships of different land plants.

Vf09G2138. The above-mentioned sequence-verified products and the GFP fragment were cloned into the pCHF3 vector by Infusion (Invitrogen) to create fusion constructions driven by the 35S promoter. The tobacco leaves were infected, and then the GFP signals were observed under Confocal Microscope (TCS-SP8 Leica, Wetzlar, Germany), according to the previous methods [32].

3. Results and discussion

3.1. The AHL genes in plants

In general, members of the AHL family have two functional domains, PPC domain and AT-hook motif [2]. To gain the universal AHLs characteristics in plants, our analysis focused on AHLs in 20 genomes: *V. fordii*, *V. montana*, *A. thaliana*, *P. trichocarpa*, *B. distachyon*, *V. vinifera*, *P. persica*, *Z. mays*, *R. communis*, *P. brechtsneideri*, *M. esculenta*, *H. brasiliensis*, *J. curcas*, *O. sativa*, *P. patens*, *S. moeileendorfi*, *M. pusilla*,

O. lucimarinus, *C. reinhardtii*, and *V. carteri* (Fig. 1a). Using both HMMER 3.0 and BLASTp [33], a total of 371 AHLs were identified from these genomes (Fig. 1a and Table S1).

In the present study, we speculated the initial phylogenetic relationship of AHLs, and further divided these AHLs into three clades: Type-I, Type-II, and Type-III, which was basically consistent with the previous results [1,8]. According to these results, we proposed a hypothesis evolutionary relationship of AHLs (Fig. 1b). In our hypothesis model, we speculated that PPC gene encoded a PPC protein and existed in early Viridiplantae. Later, the AT-hook motif (H1) was added at the N-terminus to produce an AHL protein with a PPC domain and H1 motif, which was the early AHL member belonging to Type-I AHLs (H1 + PPC). However, some AHLs from the early land plants with two AT-hook motifs (H1 + H2) indicated that the first AT-hook motif duplication event occurred before lower land plants evolved into higher land plants, and these generated AHLs with double AT-hook motifs and PPC domain belonging to Type-II AHLs (H1 + H2 + PPC). A deletion event of the H1

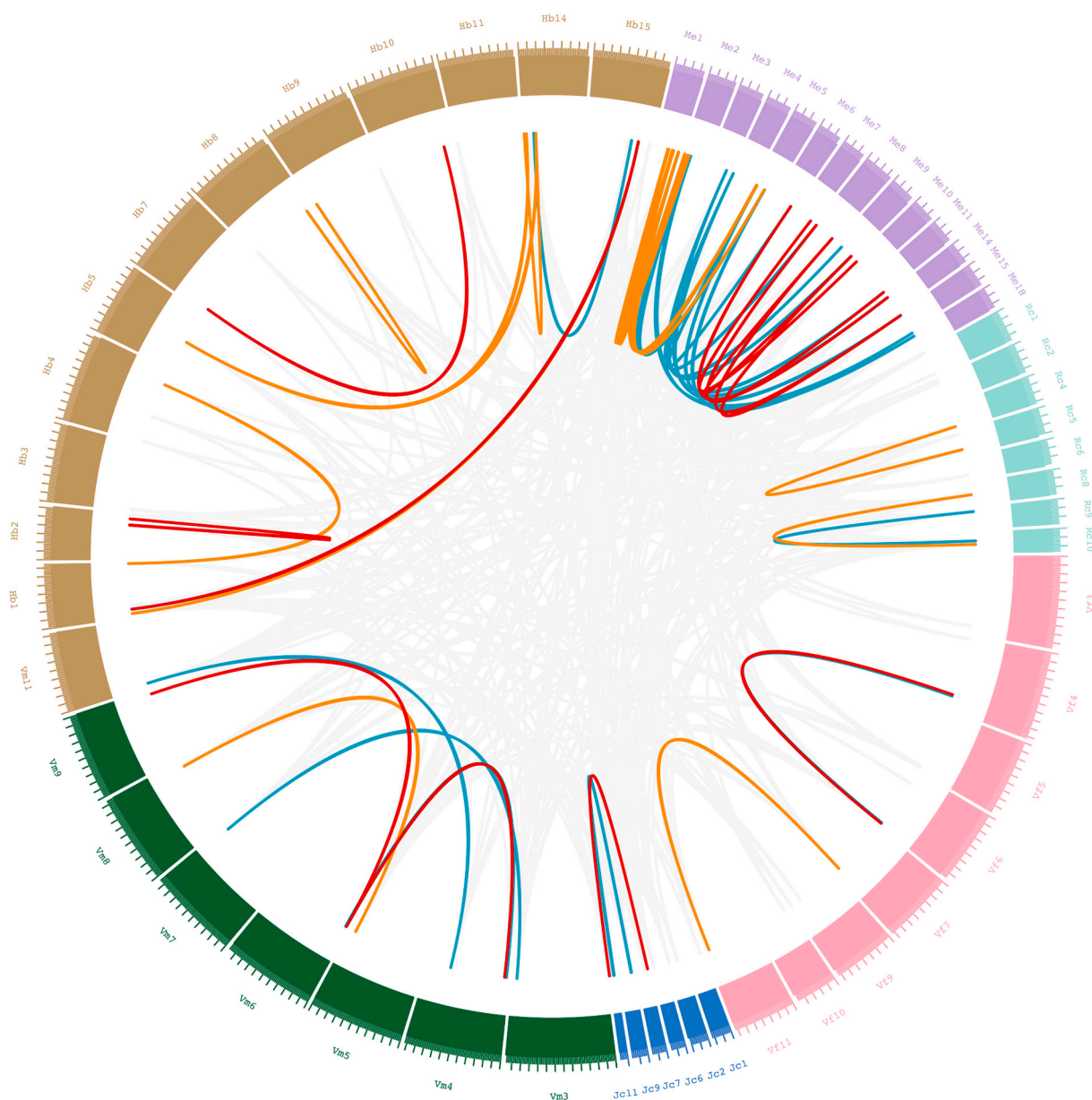


Fig. 4. Collinearity of AHLs in the Euphorbiaceae. The gray lines indicated collinearity between species. The orange, blue, and red lines represented Type-I AHLs, Type-II AHLs, and Type-III AHLs gene duplication events, respectively. Rc, Me, Hb, Jc, Vf, and Vm suggested *R. communis*, *M. esculenta*, *H. brasiliensis*, *J. curcas*, *V. fordii* and *V. montana*, respectively.

motif of an AHL belonging to Type-III AHLs rises to an AHL with a single AT-hook motif and PPC domain (H2 + PPC), a characteristic of the structure Type-III AHLs. This hypothesis model was supported by differences and similarities in these domains (Fig. 2a). For instance, the H1 motif in Type-I AHLs contained a great difference compared with this domain of Type-II AHLs (Fig. 2a). These changes in the AT-hook motifs and PPC domain led to the origin of different clades, which generally occurred in the early evolutionary stages of plant genomes with retaining the biological functions of the AT-hook motif.

3.2. Conservation and divergence of AT-hook motif and PPC domain

Members of the AHL family were conserved among land plant genomes, and all members contained one PPC domain and one or two AT-hook motif(s) [1,7,9]. According to the number and type of AT-hook motif(s), the AHL family members identified in our study could be divided into three clades, including Type-I AHLs, Type-II AHLs and

Type-III AHLs (Fig. 2b). The proteins from Type-I AHLs had a PPC domain and an AT-hook motif (H1). The proteins from Type-II AHLs contained a PPC domain and two AT-hook motifs (i.e. H1 and H2). However, the proteins from Type-III AHLs had a PPC domain and an AT-hook motif (H2) (Fig. 2).

The PPC domain of Type-I AHLs proteins displayed conserved L-R-S-H motif, while this domain of Type-II AHLs and Type-III AHLs proteins both generally had the F-T-P-H motif, which were basically consistent with the previous results [1,5,9,12]. For example, the *ZmAHLs* from Type-III AHLs contain a completely conserved F-T-P-H motif [6]. However, the divergence of PPC domain in AHLs was further followed by moderate conservation, which contains a conserved Gly-Arg-Phe-Glu-Ile-Leu motif. We also observed that the diversity of AHLs extends beyond the PPC domain sequences and exhibited in AT-hook motif(s), which contained a core sequence of R-G-R. These core sequences in Type-I AHLs were R-G-R-P, and that of both Type-II AHLs and Type-III AHLs were R-G-R-P-R-K-Y (Fig. 2). These data further confirm our

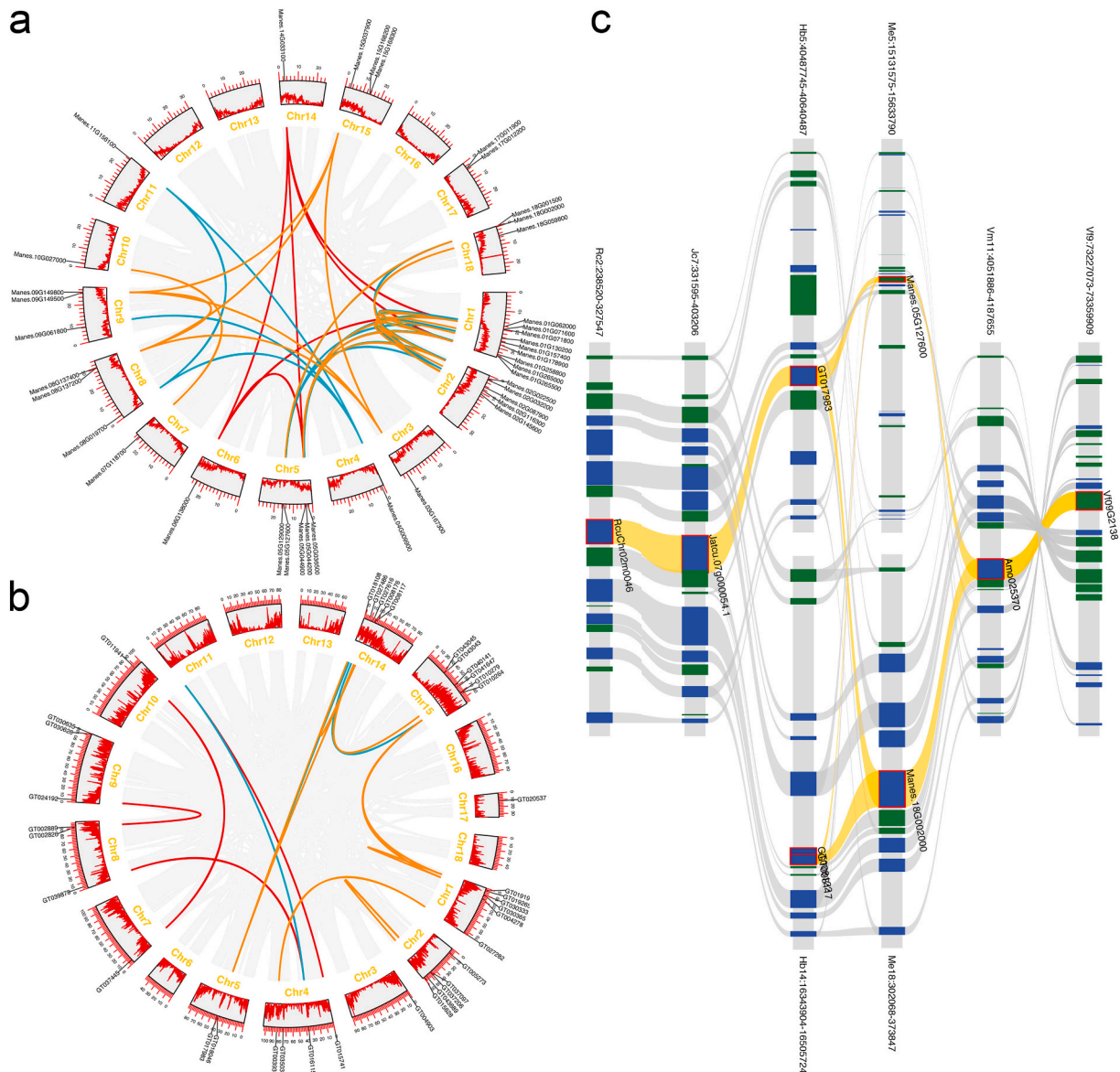


Fig. 5. Collinearity analyses of AHLs. (a) Chromosomal distribution and whole-genome duplication (WGD)/segmental duplication of AHLs in *M. esculenta* and *H. brasiliensis*. The orange, blue, and red lines represented Type-I AHLs, Type-II AHLs, and Type-III AHLs gene duplication events, respectively. (b) The *Vf09G2138* was selected for further analysis in Euphorbiaceae as an example. The yellow line indicated that they have a conservative collinearity relationship in Euphorbiaceae. Green and blue indicated the gene's orientation.

above hypothesis evolutionary model that both Type-II AHLs and Type-III AHLs evolved from Type-I AHLs (Fig. 1b).

3.3. Evolution analysis of AHLs in plant genomes

To gain insight into the evolutionary origin of the AHLs, we built ML trees and carried out collinearity analyses. Firstly, we searched the AHLs in chlorophytes by HMMER and BLASTp software. Additionally, we also used the AtAHL proteins from *A. thaliana* to search the Phytozome database. In chlorophytes genomes, we did not identify any AHL proteins (Fig. 1a), and only detected PPC protein that contained the PPC domain without AT-hook motif(s) in some genomes, such as *M. pusilla* and *O. lucimarinus* [34,35]. Remarkably, this result was also found in archaeal and bacterial genomes, in which each genome only has a single PPC protein [2].

To elucidate the evolutionary differences and origins of these AHL families in plants, the molecular history of these genes was further analyzed. Firstly, we performed maximum-likelihood (ML) analysis on

the retrieved AHLs (Fig. 3). This ML tree exhibited these AHLs were clustered into two monophyletic clades, Clade-A and Clade-B, with five (A–E) and four (F–I) subfamilies, respectively. Subsequently, we also detected the orthologous relationships with AHLs of these plants. According to the previously published articles [36], we defined the orthologous groups between AHL genes and finally identified 2150 orthologous gene pairs (Fig. 3 and Table S2). Our data highlighted that there is a correspondence between orthologous groups and phylogenetic relationships (Fig. 3). The AHLs from Rosaceae, Poaceae, Brassicaceae, Salicaceae, and Euphorbiaceae families contributed at least one AHL to each subfamily, indicating that these genes may have evolved from a subfamily-specific ancestral gene and then functional differences have occurred between these subfamilies. Especially, within each subfamily of Clade-B, the AHLs from Type-II AHLs and Type-III AHLs tended to cluster together and form Type-II AHL sub-clade (Subfamily I) and Type-III AHL sub-clade (Subfamily H), respectively. Within the subfamily G and subfamily F, most AHL genes from Type-II AHLs and Type-III AHLs could all be observed, suggesting that there is a close evolutionary

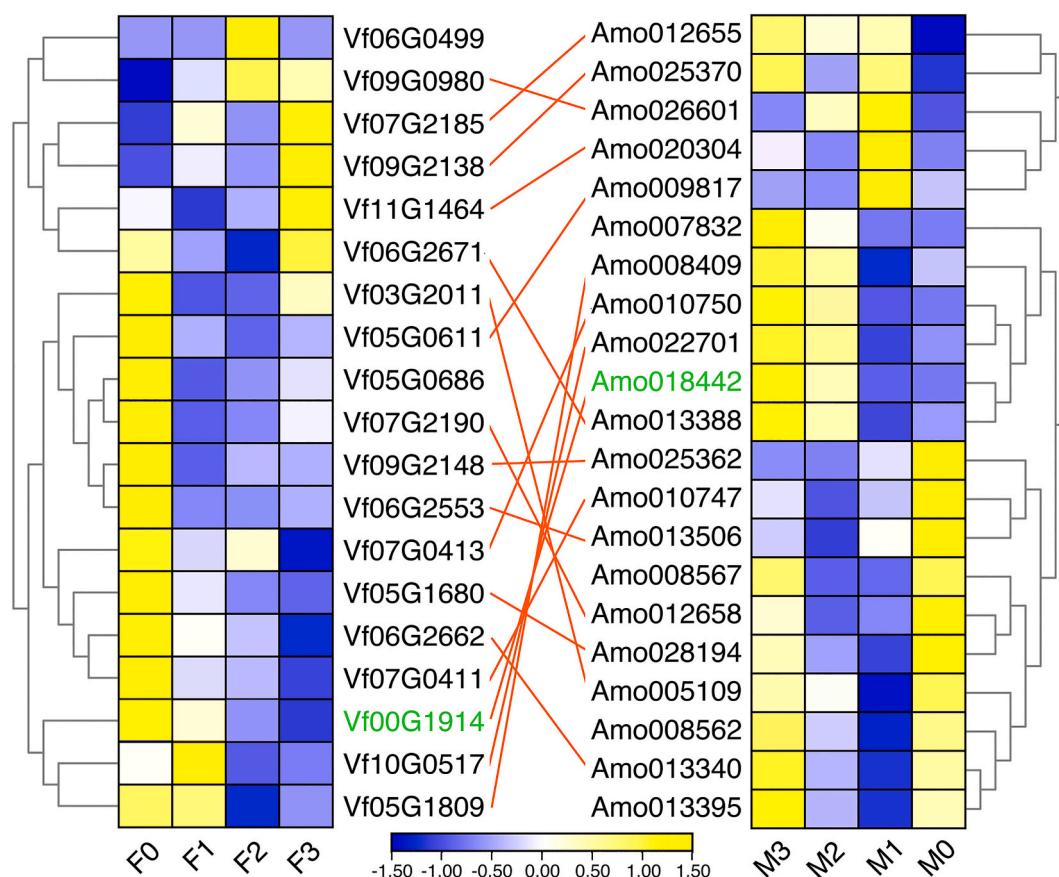


Fig. 6. Expression profiles of AHLs under infection in *V. montana* and *V. fordii*. The yellow lines indicated the one-to-one orthologous relationship of AHLs between *V. montana* and *V. fordii*. F0-F3 and M0-M3 represented the expression of AHLs in *V. fordii* and *V. montana* during the infection four stages by the pathogen *Fusarium* wilt, respectively, as follows: F0-M0, uninfected stage (0 days post-infection (dpi)); F1-M1, early stage of infection (2 dpi); F2-M2, middle stage of infection (8 dpi); F3-M3, late stage of infection (13 dpi).

relationship between AHLs from Type-II AHLs and AHLs from Type-III AHLs. The genomes of lycophyte *S. moellendorffii* and moss *P. patens* did not have any AHLs from Type-III AHLs, indicating that the loss of H1 in Clade-B occurred after lycophytes diverged from the rest of vascular plants.

3.4. Collinearity analysis of AHLs in plant genomes

To identify how AHLs evolve, we also investigated the collinearity between and within genomes. In our study, most members of the AHL family could be distributed on chromosomes, except for those of *S. moellendorffii*. The collinearity pairs of all genes were determined by MCScanX. We considered the adjacent genetic relationships between *R. communis*, *M. esculenta*, *H. brasiliensis*, *J. curcas*, *V. fordii*, *V. montana* and all members of Euphorbiaceae (Fig. 4). Therefore, a collinearity alignment analysis was further performed for these six genomes. We found that these collinear pairs of AHLs were not linearly relational, according to the number of AHLs within a species. Further, the number of duplicated gene pairs did not seem to be affected by the genome size and chromosome number, as shown in *A. thaliana* and *B. distachyon*. Although the genome size and chromosome number were similar, the number of collinear gene pairs varied among these genomes (Fig. S1 and Table S3).

The analysis of duplicated events suggested that the whole genome duplication (WGD)/segmental duplication was the major AHL duplication type in studied all genomes except *P. patens*, and *S. moellendorffii*, indicating that most of AHLs were retained after WGD/segmental duplication in plant genomes (Fig. S1 and Table S3). In *P. patens*, which has undergone one or several WGD [37,38], most AHLs in these plants

were produced by single gene duplication, which might be affected by the frequent activities of transposons. The number of AHLs in *S. moellendorffii*, which has not experienced WGD [39,40], was less than that of AHLs from other plant genomes. Indeed, the AHL evolved from an ancestor with highly conserved in Euphorbiaceae, and the WGD events promoted the differentiation of species in this family and increased the proportion of AHLs in both *M. esculenta* and *H. brasiliensis* genomes (Fig. 5). These data indicated that WGD played an important role in the expansion of AHL family members in plants.

3.5. Expression profiles of AHLs in *V. montana* and *V. fordii* during infection

V. fordii is a promising oil-producing species. Unfortunately, the devastating *Fusarium* wilt disease has a great adverse effect on its yield and growth, while its sister species (*V. montana*) was found remarkably resistant to *Fusarium* wilt disease [21,22,41,42]. Several studies have been found that members of the AHL family are widely involved in plant stress resistance [18,43–45]. For example, overexpression of *CaATL1* (*Capsicum annuum* AHL gene) in tomato significantly enhanced its resistance to oomycete and bacterial pathogens [18]. Therefore, AHLs might be candidate genes for resistance to *Fusarium* wilt in future genetic breeding. To determine the expression levels of AHLs under *Fusarium* wilt infection, the expression patterns were implemented at four infection phases in *V. montana* (M0, M1, M2, and M3) and *V. fordii* (F0, F1, F2, and F3). As shown in Fig. 6, both down-regulated and up-regulated expression patterns were reflected in AHLs in these two *Vernicia* species. Twelve AHLs from *V. fordii* and six AHLs from *V. montana* exhibited down-regulated expression during pathogen infection, respectively.

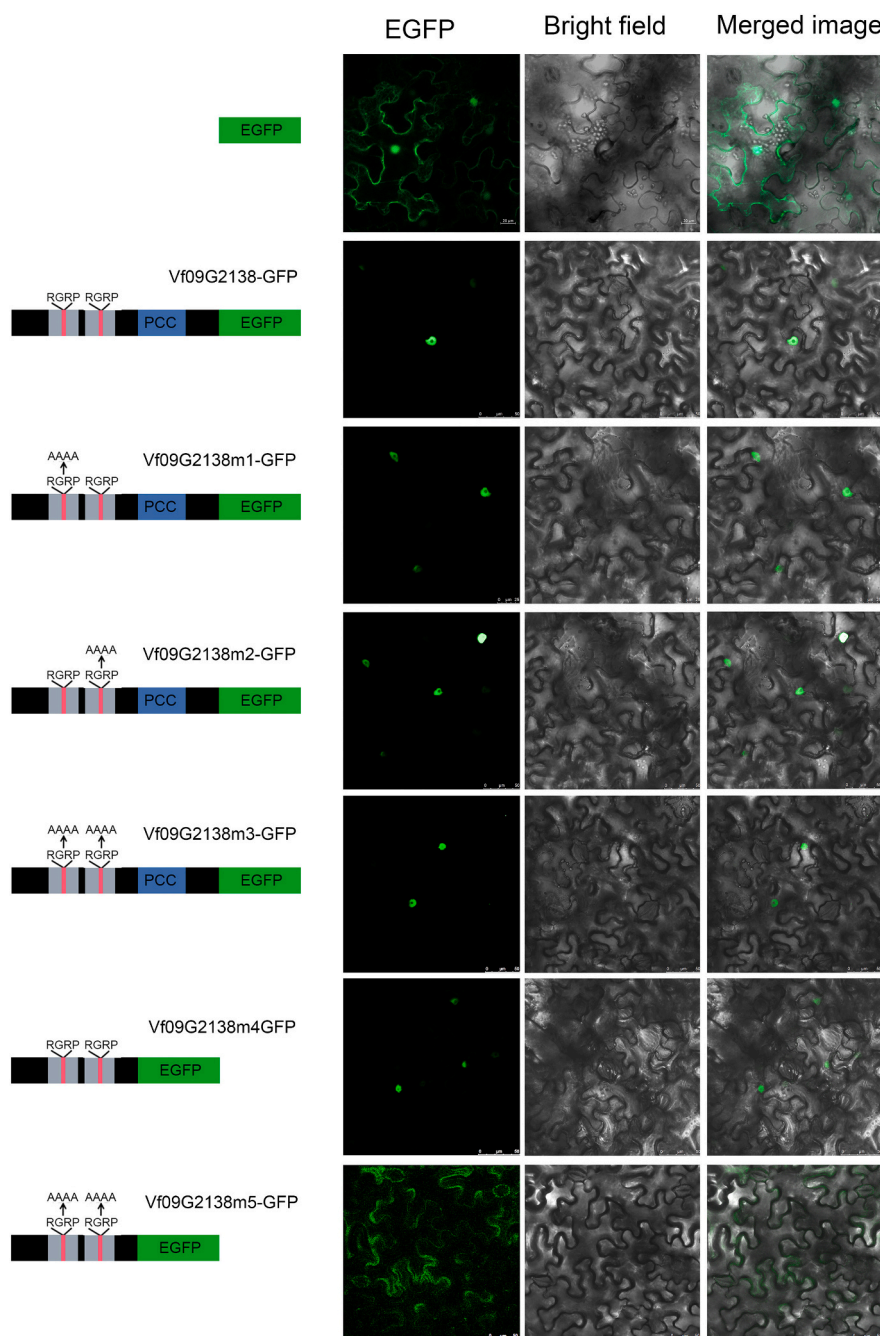


Fig. 8. The role of AT-hook motif(s) and PPC domain on nuclear localization of Vf09G2138. The positive control was EGFP driven by the CaMV35s promoter. The PPC domain and AT-hook motif were indicated blue and gray box, respectively.

amino acids with one PPC domain and two AT-hook motifs. To further understand the functionals of PPC domain and AT-hook motif, several over-expression vectors were constructed that encode one of five types of mutant *Vf09G2138* or a full-length *Vf09G2138*. The four core amino acids of H1 in mutant 1 protein were replaced by alanine (from RGRP to AAAA), and contained H2 and PPC domains; The four core amino acids of H2 in mutant 2 were replaced by alanine (from RGRP to AAAA), and contained H1 and PPC domains; The four core amino acids of both H1 and H2 in mutant 3 protein were replaced by alanine (from RGRP to AAAA); The PPC domain of mutant 4 protein was deleted and contained H1 and H2; On the basis of mutant 4, the four core amino acids of both H1 and H2 in mutant 5 protein were replaced by alanine (from RGRP to AAAA). As shown in Fig. 8, the full-length Vf09G2138 (AHL-GFP), mutant 1 (Vf09G2138m1-GFP), mutant 2 (Vf09G2138m2-GFP), mutant

3 (Vf09G2138m3-GFP), and mutant 4 (Vf09G2138m4-GFP) proteins were specifically localized in the nucleus, whereas the mutant 5 (Vf09G2138m5-GFP) as found to distribute throughout the cell. These results suggested that the PPC domain and core AT-hook motif and/or the entire region after the PPC domain in Vf09G2138 could be led Vf09G2138 to the nucleus.

4. Conclusions

In our study, we identified 371 AHLs in multiple plants, varying from the early diverging lycophyte *S. moellendorfi* and moss *P. patens* to a variety of higher plant genomes. Our analyses suggested that members of the AHL family can be divided into two clades, in which AHLs from Type-I AHLs differentiate in Clade-A, and AHLs from both Type-II AHLs

and Type-III AHLs differentiate into Clade-B. Importance of *AHLs* in economically important indicators including growth and development, and defense was evident from studies in model plants. According to the RNA-seq data, the functions of *AHLs* were related to the inhibition of seed oil accumulation, and resistance and defense against *Fusarium* wilt disease in *Vernicia* species.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2022.03.130>.

Data accessibility

Expression data of *Vernicia fordii* and *V. montana* were available in NCBI SRA database with accession numbers: PRJNA318350, PRJNA483508, and PRJNA445068.

CRediT authorship contribution statement

YC, YL, WM, LW, and LZ designed and performed the experiments. YC and LJ analyzed the data. YC contributed reagents/materials/analysis tools. YC wrote the paper. All authors contributed to the article and approved the submitted version.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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